JIHANE BENZERKANE

4th-Year Engineering Student, Digital Transformation & AI (TDIA) | ENSAH, Université Abdelmalek Essaâdi, Morocco
benzerkane.jihane@etu.uae.ac.ma | +212 644-169773
linkedin.com/in/jihanebenzerkane / github.com/jihanebenzerkane / f1track.duckdns.org

Profile

Fourth-year engineering student in Digital Transformation & AI with a strong interest in **scientific computing, simulation and large-scale data systems**. Creator of **F1Track**, a live telemetry and Monte Carlo-based prediction platform deployed on Oracle Cloud Infrastructure. One of 12 students **selected nationwide** for **Oracle's** R&D Summer Bootcamp. Seeking a **research internship at CERN** to apply my software, data and algorithmic skills to scientific computing.

Research-Relevant Projects & Experience

F1Track: Live Formula 1 Analytics & Prediction Platform (Master Project) | Feb 2026 - Present | f1track.duckdns.org

- Live telemetry cockpit (speed, RPM, throttle, brake) streamed from the OpenF1 API and refreshed every 500 ms.
- Predictive analytics layer built on Monte Carlo simulation, alongside an interactive 3D explorer of 6 team cars.
- Historical Formula 1 database (1950–2026) of 185,000+ records across 33 tables, served through 20+ REST APIs.
- Deployed with Docker, Nginx and SSL on Oracle Cloud Infrastructure (OCI).

Stack: Java, Spring Boot, React, Three.js, SQL, Docker, Nginx, OCI

PFE Assignment Platform | Automated data migration & allocation | Feb 2026 – Jun 2026

- Automated import of multi-sheet Excel files into a MySQL database, with validation and audit logs.
- Constraint-based scheduling algorithm assigning 105+ students to supervisors, balancing workloads and avoiding conflicts.
- Administrative dashboards and automated PDF reports.

Stack: PHP, Laravel, MySQL

Software Engineering Intern (hybrid) | ECS Informatique, Rabat | July - August 2026 (2 months)

- Built Raven, a maintenance-visit planning platform (C#/ASP.NET Core, SQL Server) with a priority-scoring algorithm, technician-assignment recommendations and an Excel/PDF import-export pipeline.

Honors & Selected Programs

Oracle Morocco R&D Center Summer Bootcamp (FY27) | Casablanca, Morocco | Summer 2026 | July 2026

- Selected as 1 of 12 student representatives from universities and engineering schools nationwide.
- Immersed in Oracle's R&D environment through masterclasses and technical sessions led by its researchers on microservices, cloud infrastructure, and distributed computing.

Education

Engineering Diploma in Digital Transformation & AI (TDIA) | *2023 - 2028 (Expected)*

ENSAH (Université Abdelmalek Essaâdi) - Currently in Year 4 (Bac+4) of a 5-year integrated state engineering degree.
Grades: 2023–24: 14.45/20 (3.61/5) | 2024–25: 13.43/20 (3.36/5) | 2025–26: 14.32/20 (3.58/5), all years validated.

Technical Competencies

Languages & Core Tech: Python (FastAPI), C/C++, Java (Spring Boot), C# (.NET / ASP.NET Core), SQL (Oracle SQL, MySQL, SQL Server), JavaScript (React, Three.js), PHP (Laravel).
Scientific & Data Computing: Monte Carlo Simulation, Predictive Analytics, Algorithmic Scoring, ETL Pipelines, Constraint-Based Optimization.
Cloud & Tools: Oracle Cloud Infrastructure, Docker, Nginx, RESTful APIs, EF Core, Git.

Certifications

- Oracle OCI AI Foundations Associate | Oracle | Feb 2026
- Oracle Generative AI Professional | Oracle | 2026
- Oracle Java SE 21 Developer | Oracle | 2026

Activities & Languages

Activities: President, ENSAH Chess Club (2024 – 2026)|Graphic Designer & Event Organizer, Enactus ENSAH (2023 – 2025)
Languages: Arabic (native) | French (fluent) | English (B2) | Italian (beginner)